

a larger CCD means better image quality. The smallest starts at 1/6 inch. These are usually found in lower-priced and budget camcorders. Better models have 1/3 inch and 1/4 inch CCDs. The difference in CCD sizes actually becomes apparent when you compare picture brightness and colours. Larger CCDs will give better picture quality and more vibrant colours. A larger CCD will also give you better quality in low-light situations.

When picture quality really matters, a 3-CCD camcorder should be considered rather than a single-CCD. This was the exclusive realm of professional video cameras once, but is now increasingly found in mid-range models. In 3-CCD camcorders, there is one CCD for each of the primary colours. Colours on 3-CCD camcorders are brighter, more vivid, life-like, and accurate.

Finally, take into account the number of pixels on the CCD. A higher pixel count indicates sharper images. However, don't be fooled by the manufacturer's claim of having a higher *gross* pixel count. The keyword is *effective* pixel count. The effective count is what the CCD actually uses. Low-end budget camcorders should have around 290,000 pixels or higher effective. Standard video has around 340,000 pixels effective. Beyond the 340,000 mark, the increase in pixel count has little effect on sharpness, and is subject

CMOS Sensors

While CCDs will remain a staple feature of the camcorder for some time to come, advances in CMOS imaging technology are finally enabling the appearance of CMOS-based camcorders. CMOS imaging sensors are currently used in very high end digital cameras and low-end webcams, but used to suffer from more noise (grainy pictures) than CCDs of comparable performance. These issues are being overcome, and the digital imaging industry is set to go in the direction of CMOS imaging sensors. CMOS sensors have some advantages over CCDs. Both use pixels on which light falls, which is converted to electrical energy and then processed. The difference between CMOS and CCD is that the processing on CMOS is done at the pixel itself, while in CCDs the electrical signal is sent to separate chips for processing.